

Individual Reflection on Group Project

Project Title: Edge-Deployable Rock Detection and Burial Status Classification

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Team / Group: Team 9

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1. Your views on the completing the project as a team

Working as a team was valuable because the project combined data preparation, model development, edge deployment and academic reporting. My responsibilities were dataset preparation and splitting, image augmentation, YOLO augmentation control and training configuration, YOLOv8n and YOLO11n training and evaluation, cross-model comparison, and Tableau visualisations for the final report. Dividing these tasks allowed parallel progress through shared datasets, checkpoints and metrics. Because my outputs fed later analysis, an incorrect split or inconsistent configuration could have affected the whole team's conclusions. I therefore learned that sharing traceable results and explaining limitations were as important as training the models.

2. Your views on project management and execution of your team

Our execution followed a clear dependency order. We audited 448 images and 706 annotations, then created the 313/45/90 train-validation-test split before augmentation. I generated six photometric variants only for training, expanding that partition to 2,191 images while leaving validation and test data unchanged. A common schedule and controlled YOLO settings made the model comparison meaningful, while Tableau figures communicated the evidence concisely. I learned that effective management requires agreed evaluation criteria, consistent file naming and early confirmation of authoritative results when several people contribute to one report.

3. Reflections on Successes, Failures, and Retrospective Changes

- **What Worked Well:** Splitting before augmentation prevented leakage, and a shared configuration made the comparison fair. The predefined validation mAP50-95 criterion supported selecting YOLOv8n at 0.4288, although other models led on secondary metrics. Tableau made these trade-offs easier to interpret.
- **What Did Not Work Well:** The modest dataset and difficult half-buried class limited generalisation. YOLOv8n fell from 0.4288 validation mAP50-95 to 0.3384 on the test set. Offline and internal YOLO augmentation also required careful control to avoid an overly aggressive pipeline.
- **What I Would Do Differently:** I would define the experiment matrix and visualisation structure earlier, run an augmentation ablation, and use repeated stratified splits or cross-validation where time allowed. Earlier joint reviews would also keep metrics, labels and report wording consistent.

4. Any other reflections

This project strengthened my ability to take ownership of an end-to-end experimental stream while contributing to a shared research outcome. I became more critical about data leakage, reproducibility and the difference between selecting a model by a declared criterion and calling it universally “best.” Most importantly, I learned that rigorous analysis includes reporting weaknesses, such as the half-buried recognition gap, rather than presenting only favourable results. These lessons will influence how I plan, document and communicate future machine-learning projects.